Drive Drilling
09/05/2019

Parameter Of Drive I

*Obtaining desired shape, size, orientation & gradient of mine opening.

*Achieving accurate counter of drive (opening) with least over break & smooth surface & face.

*For compact heapining of blasted muck after blasting at working face.

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Drilling & Blasting I

*Solid blasting (blasting with out any cut) is applicable for any type of rock.

*Different drill hole pattern is followed for drilling holes at face.

*Different pattern of hole breaks rock in different way.

*In blast holes, different blast hole serves different purpose.

*Cut hole, holes which are used to create initial free face at face & subsequent holes blasts towards curt hole.

*Easers holes are located immediately close to cut hole.

Drilling & Blasting II

*Easers holes can be are slightly oriented toward cut hole.

*Easers holes provide increase excavation at mine face.

*Trimmer holes are drilled out side of eases hole.

*Trimmer expands blast / drove size.

*Contour holes are drilled at extremely out side, thesis holes are also called line, rib, side, flank holes.

*Contour holes are to maintain final size & shape of drive excavation.

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Parameter Of Drive III

*Contour holes are drilled slightly out ward to accommodate drill machine for next round of drilling.

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Different Type Of Drilling I

*Drill patterns can be derived in two groups

*<u>Parallel hole cut</u>

*Parallel holes cuts are of this type

*Burn cut.

*Helical spiral cut.

*Coroment cut

*Incline hole cut.

*Incline cut holes can be of following pattern based on shape of cut.

<u>*Wedge cut.</u>

<u>*Fan cut.</u>

Different Type Of Drilling I

*Pyramid cut.

*Drag cut

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Selection of Drill Patern I

*Different cut & drill patterns are selected based on following.

*Type of equipment deployed for drilling.

*Shape of face & drive.

*Size of face & drive.

*Level of mechanization.

*Geo technical parameter of rock.

*Bedding plan inclination.

*Joint plan pattern etc.

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Parallel Hole I

*All holes, including cut holes are parallel & at right angle (perpendicular) of drill face.

*Cut holes are normally drilled at center of face.

*Incase last blast has yielded subsential socket at face, location of burn cut is changed to get better place for drilling.

*In burn hole, some holes are charged & some holes are left empty (uncharged with explosive).

*Uncharged holes are normally reamed to larger diameter.

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Parallel Hole II

*Shock waves, when reflected at these empty holes, rock is shuttered & subsequently blown out by escaping gases.

*Three is specific geometrical relation between diameter of empty holes & space between empty hole & charged hole.

*A particular diameter of reamed hole & distance form charged hole gives optimal result.

*With spacing less than 1.5 times of diameter of empty holes, blasting is expected to be clean, while higher spacing breakage tends to be uneven.

Parallel Holes III

*With spacing in excess of twice diameter of empty hole, there is a likely hood of plastic deformation & burning or rock.

*For achieving greater advantages it is there fore necessary to increase diameter of empty holes.

*Increasing diameter of empty hole shall give scope to increase distance between holes with out jeopardizing condition of future breakage.

*Even with proper burden, if charge concentration is too high, a miss function of cut may result due to rock impact & sintering.

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Parallel Holes IV

*This shall prevent generation of planned free face.

*There are variation in shot hole pattern employing large diameter empty holes or providing empty space by not blasting few shot holes.

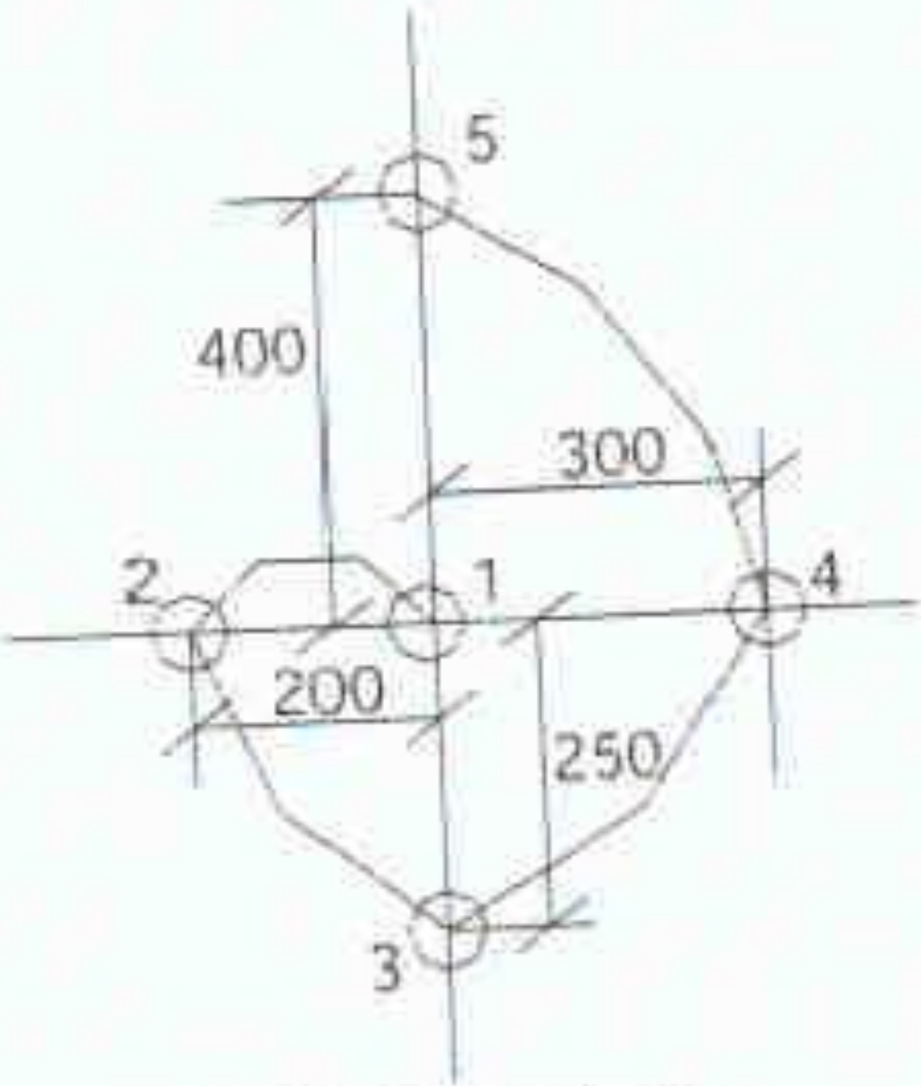
*Introducing large diameter free hole & large distance between charge hole & free hole reduces scattering of rock at face from 15-23m to 9-10 meter.

*This increases loading efficiency by about 15%.

*This pattern is suitable for hard, brittle & homogeneous rock.

*This method is also known as fragmentation or fracture cut.

Parallel Holes I



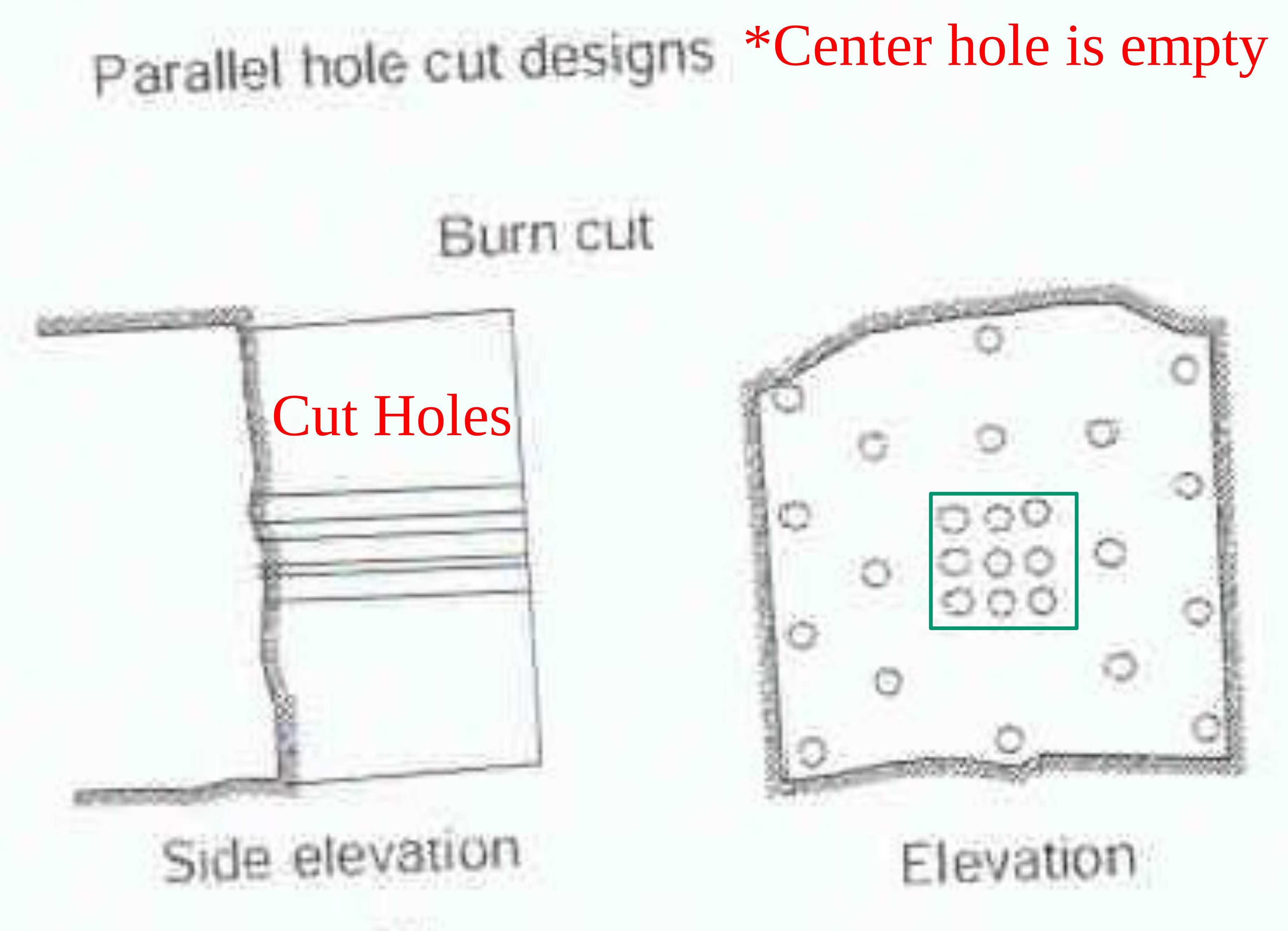
Helical spiral cut

*Helical spiral Cut

1 – free hole

2-5 delay sequence

Parallel Holes I



Parallel Holes I

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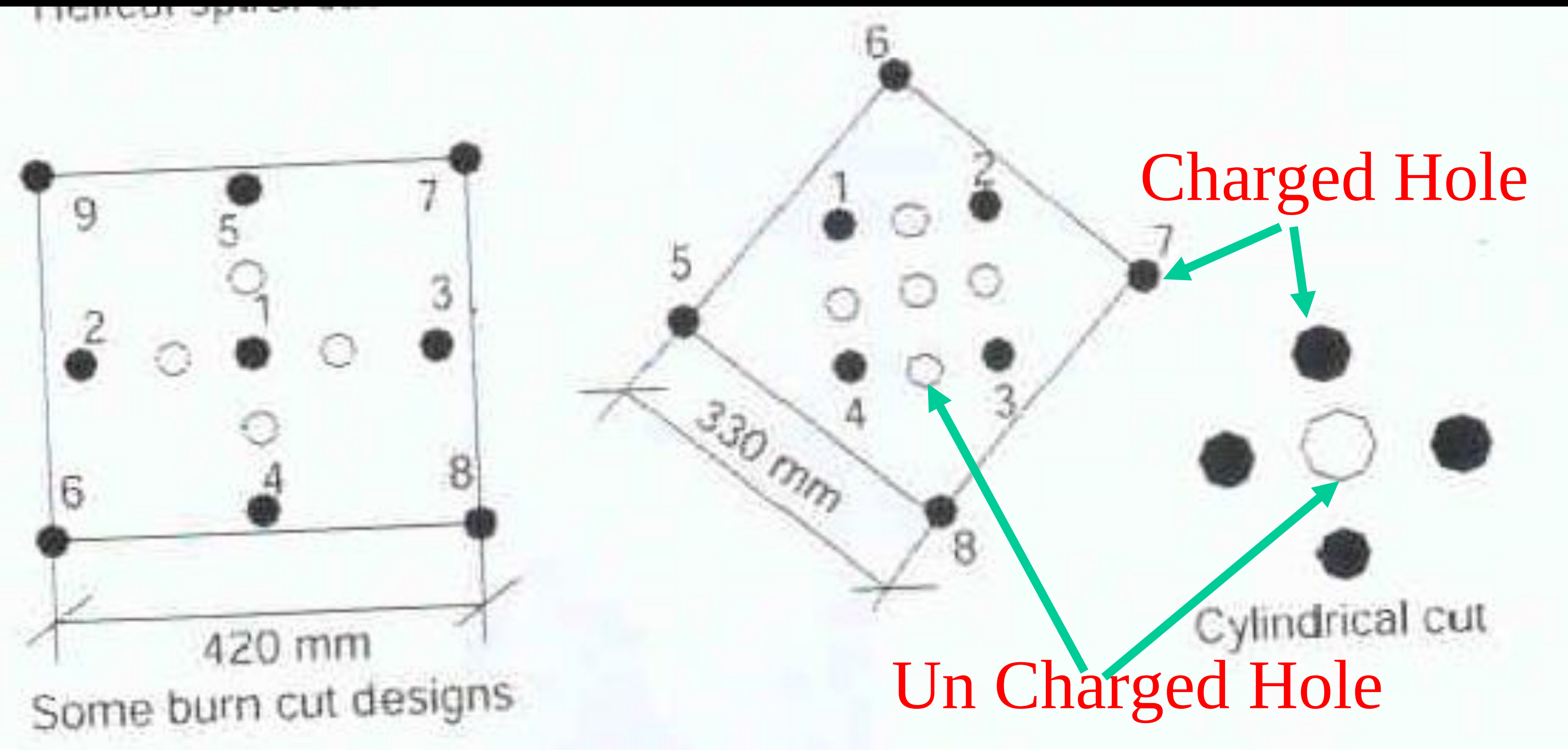
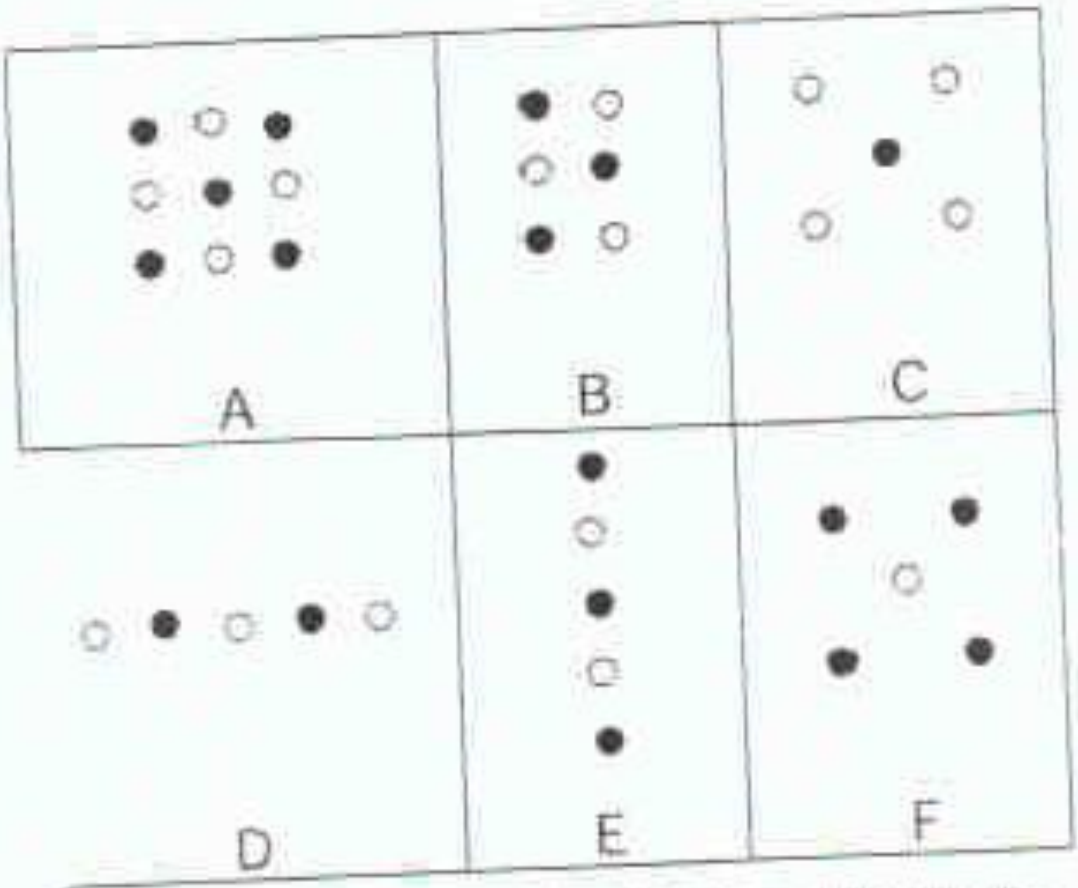


Parallel Holes I



A, B, C = Box cuts. D, E = Line cuts
F = Large uncharged centre hole
● = Charged holes
○ = Uncharged holes

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Parallel Holes I

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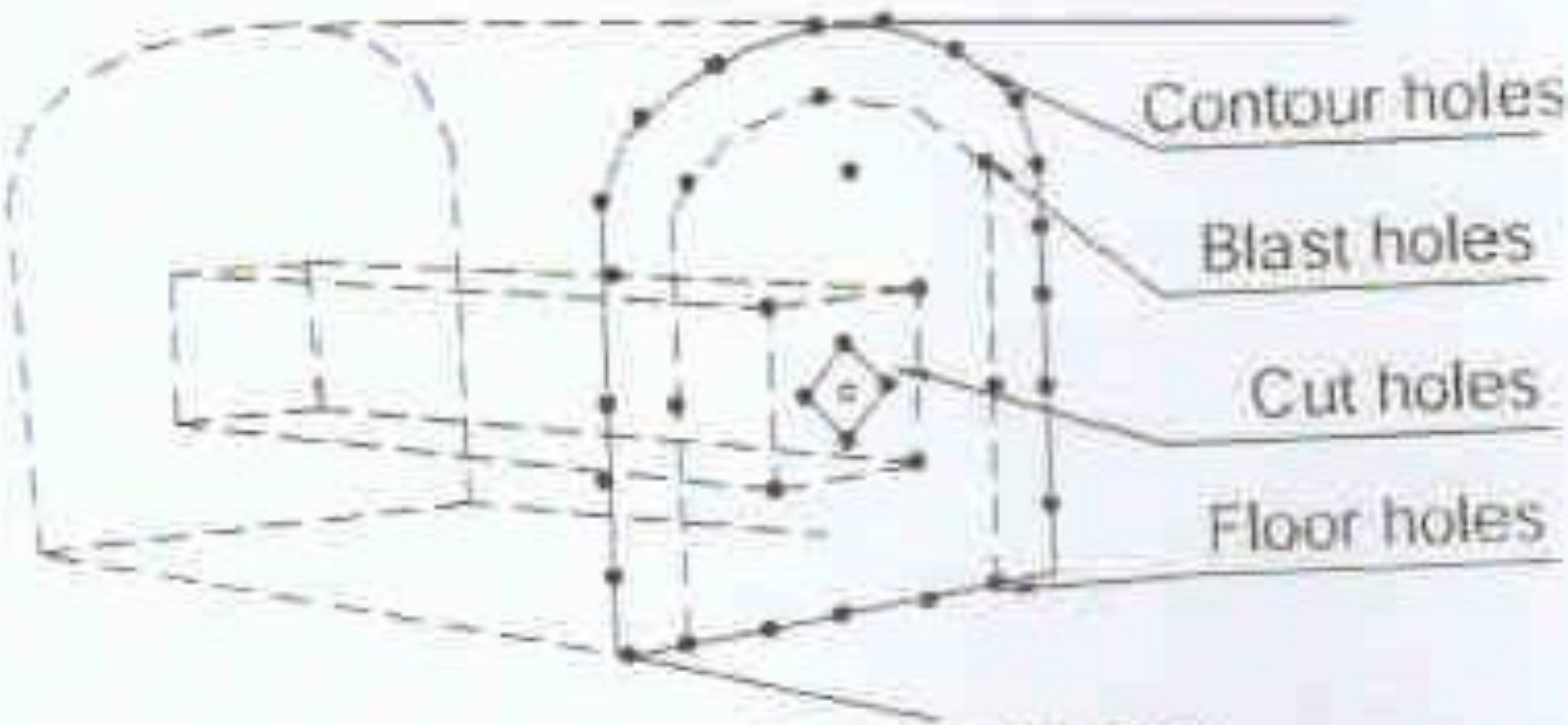
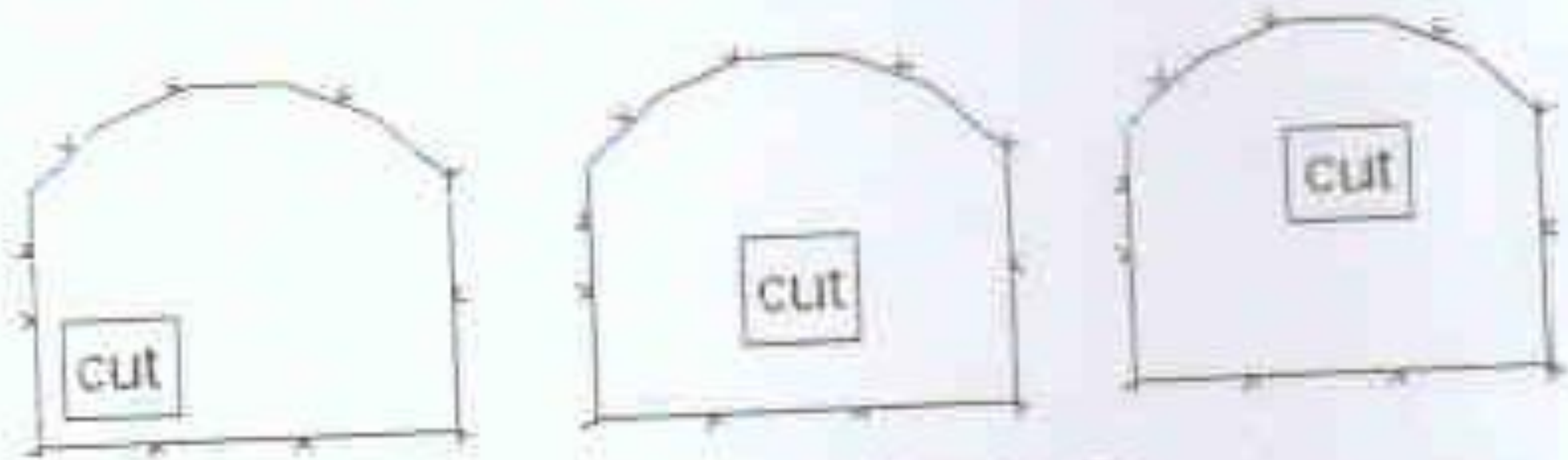


Types of holes in a tunnel face

Permanent Drive Support I

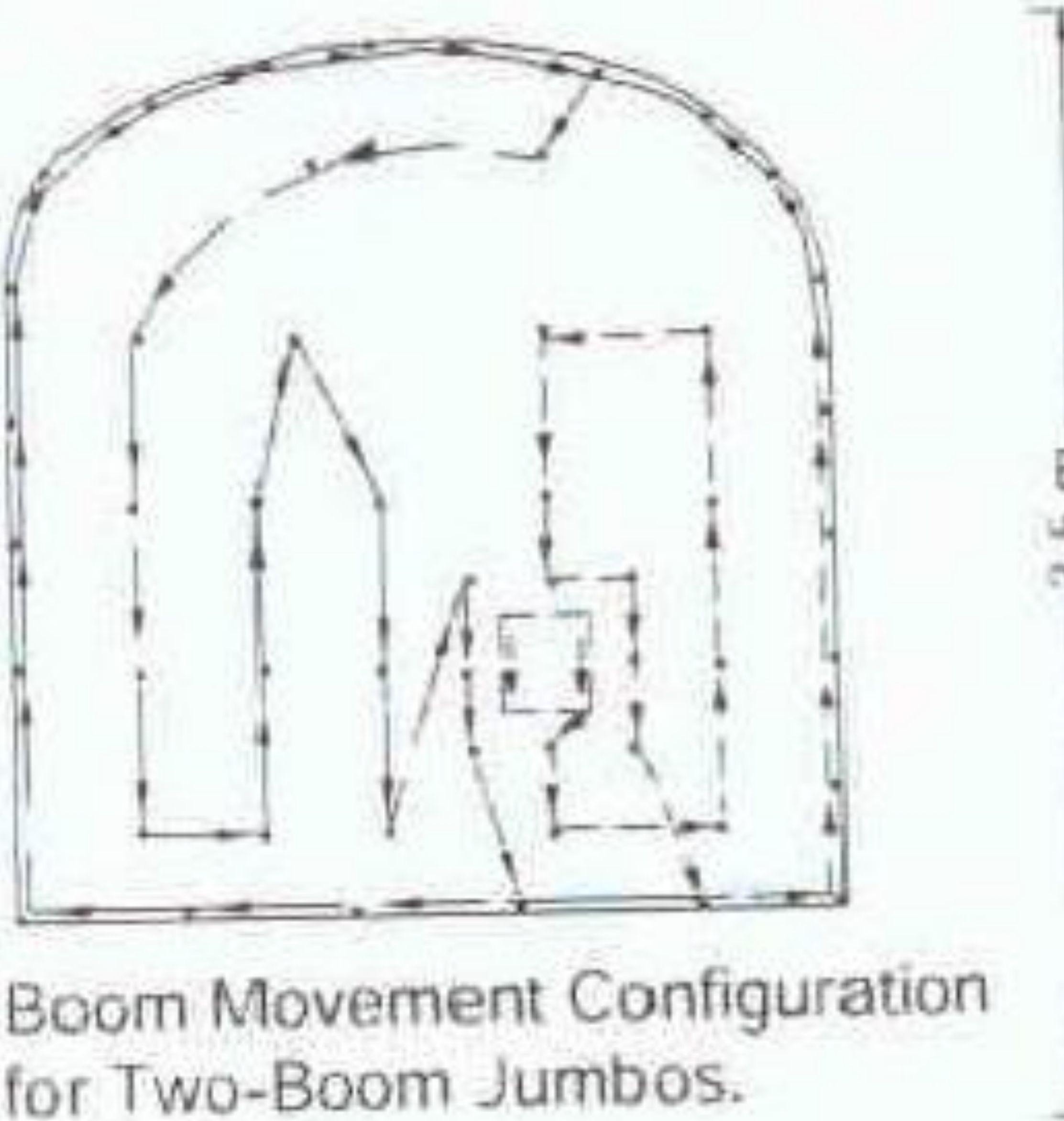
Cut holes can be located at different places in face.

This is done to avoid disturbed are at face, specially by socket of holes, left from last blast.



Various cut locations

Parallel Holes I



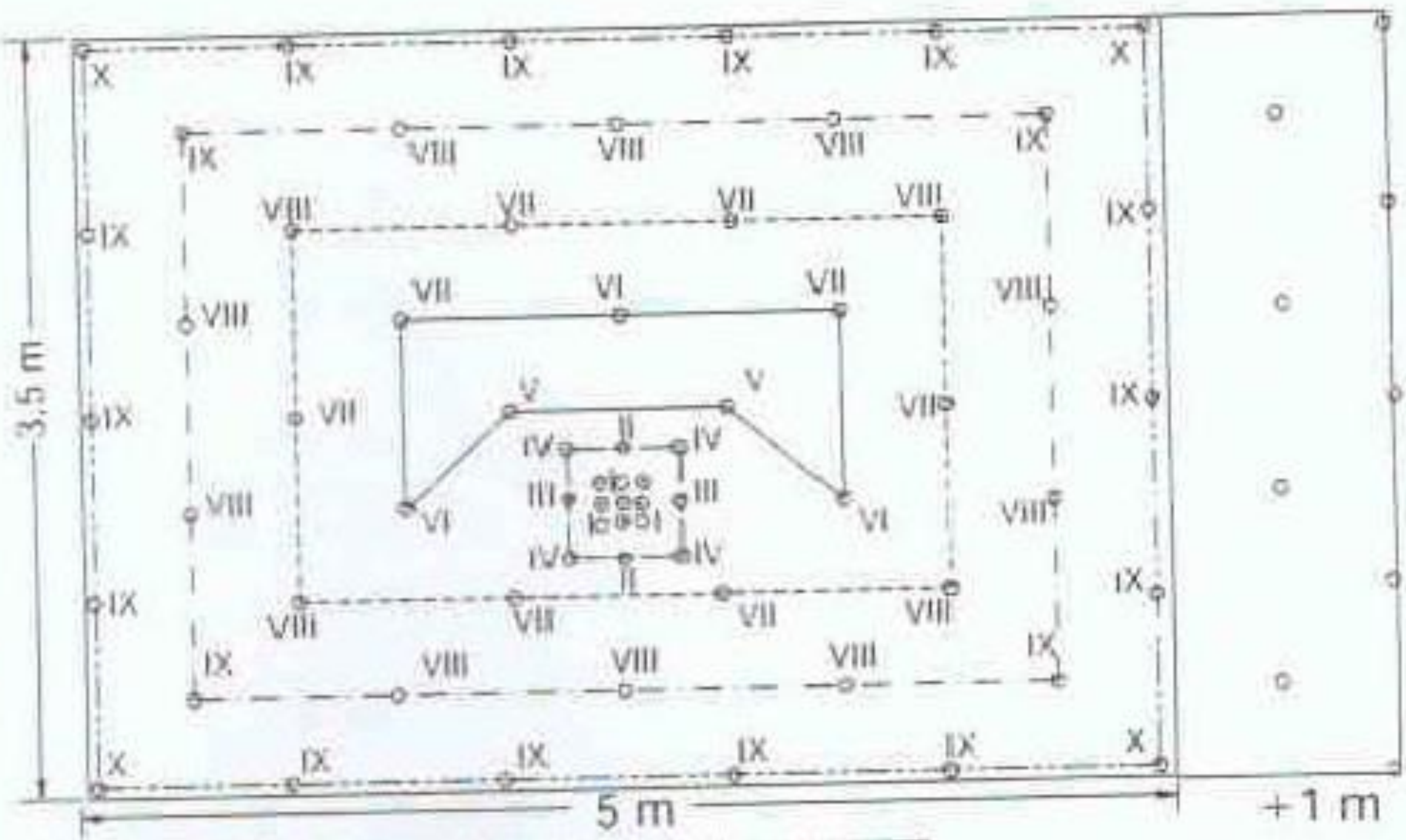
(b) Boom Movement Configuration for Two-Boom Jumbos.

When drilling is done by two booms, two booms are driven parallelly in staggered fashion. One boom is at collaring phase other boom is on drilling mode.

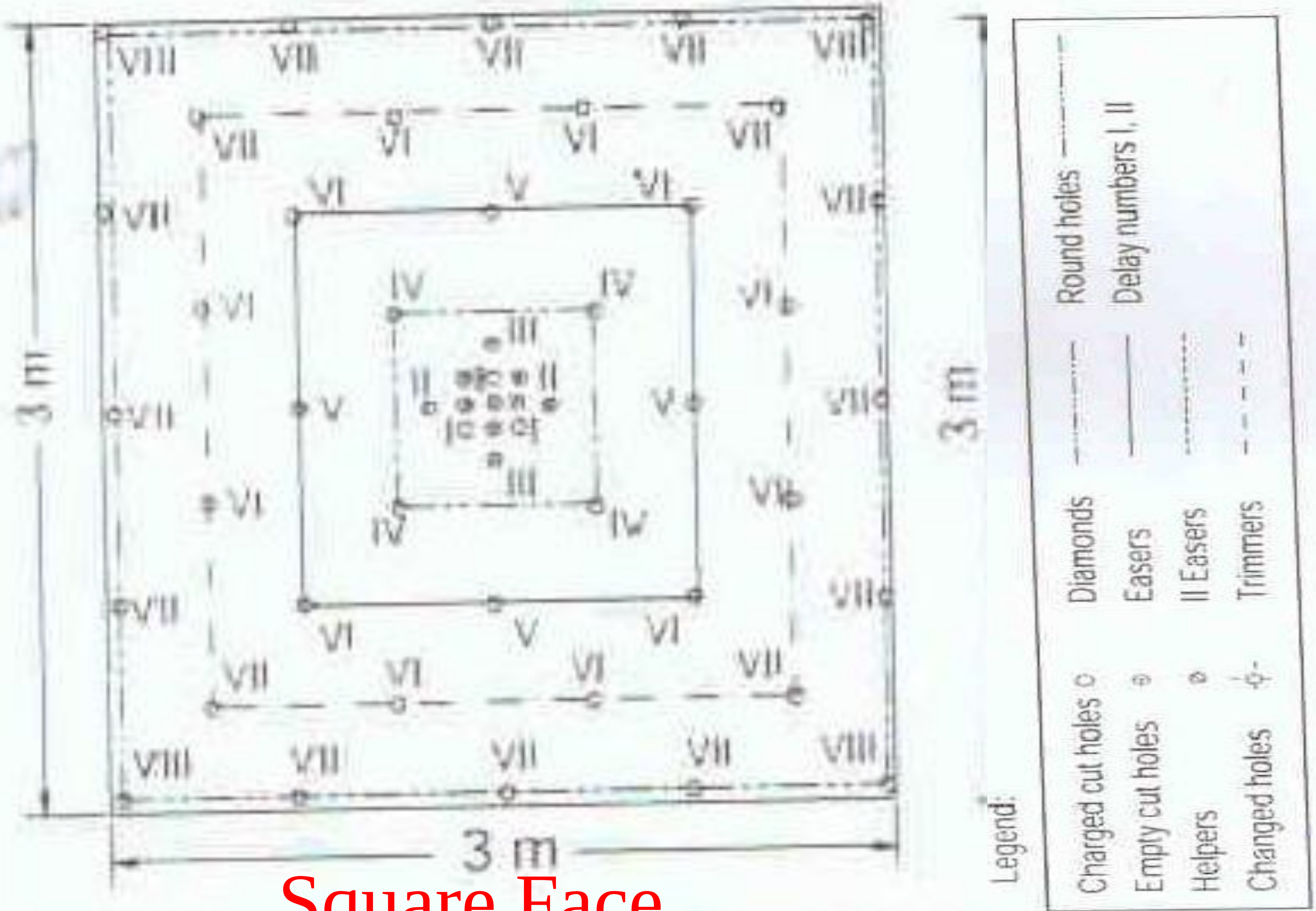
Parallel Holes

Legend:

Charged cut holes	○	Diamonds		Round holes	
Empty cut holes		Easers		Delay numbers I, II	
Helpers		II Easers			
Changed holes		Trimmers			

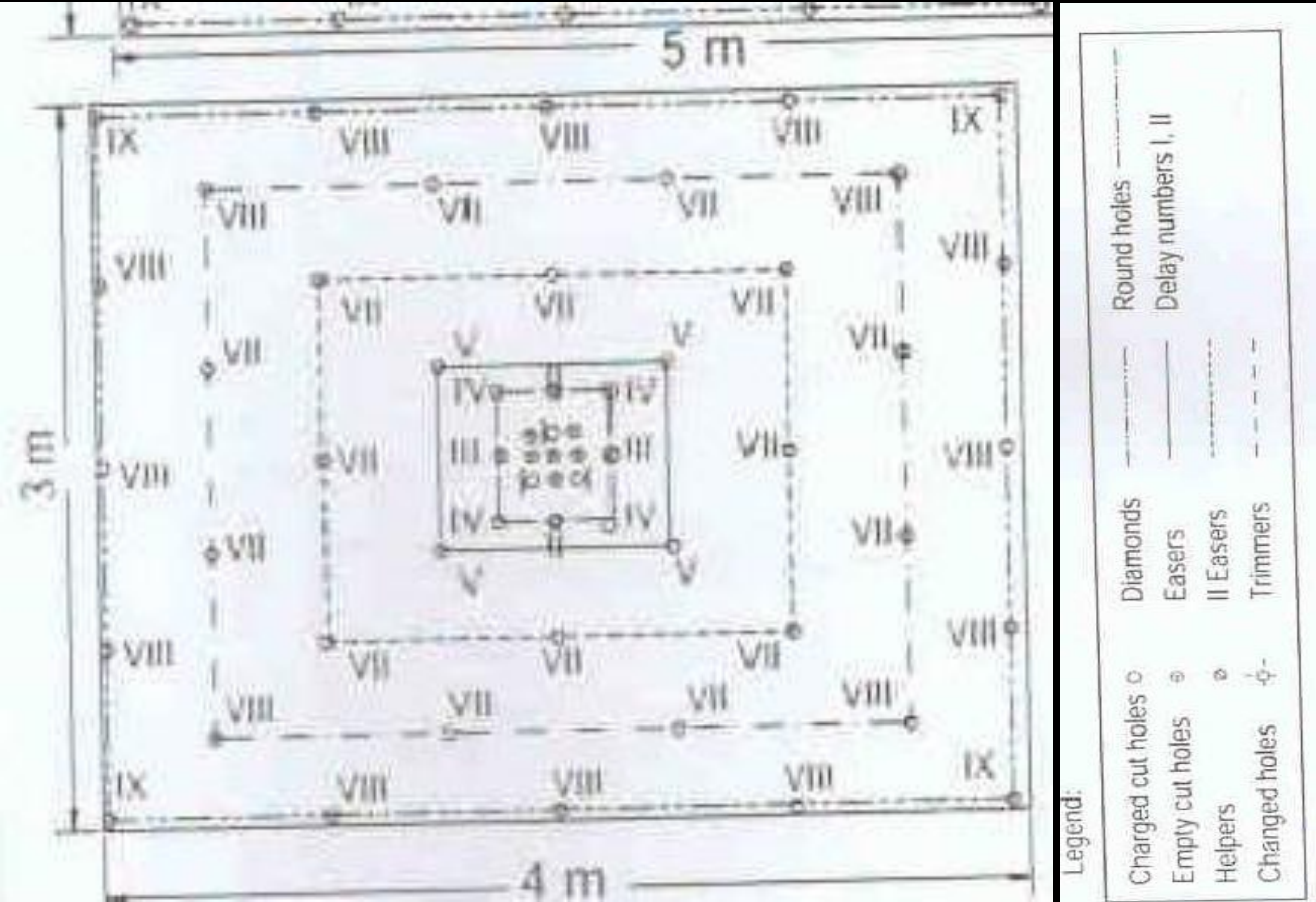


Parallel Holes



Square Face

Parallel Holes



Hole Diameter I

*Hole diameter is function of explosive cartridge to be charged.

*Difference between diameter of hole & cartridge should be small.

*Air gap reduces effect of explosive.

*Optimally drill hole should not be more than 1.2 times of cartridge diameter.

*Effect of decoupling is considerable in blasting.

*Number of hole in a round should decrease in proportion to increase in diameter of holes.

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Cut Hole I

*Location of cut hole & its design within drilling pattern meant for drive excavation is vital.

*Good cut hole design should consider following

*Diameter of large empty holes / holes.

*Burden.

*Charge concentration.

*Drilling precession & charging skill.

*Larger empty hole diameter, better it is & deeper could be depth of round.

*Slight deviation in cut hole drilling can jeopardies complete blast.

Cut Hole II

*Too large a burden will cause breakage only or plastic deformation in cut area.

*Right burden will result in a clear blast up to planned length (depth of round).

*This shall ensure better advance per round.

*Hole closet to empty hole must be charged very carefully.

*Too low concentration of charges may not break rock, while too high concentration may recompacted rock.

*This processes known freezing & there by not allowing rock to blow out through large hole.

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Cut Hole III

*Number squire in cut is limited by the fact that burden in each squire must not exceed burden of stoping holes for given charge concentration.

*Cut hole occupy approximately 2m sq are of face.

*Small drive consist only cut hole & contour hole.

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Drive Other Then Cut Hole I

*Easer holes – where burden of hole gradually increases to maximum stope burden.

*Trimer holes – holes with burden of stoping holes to increase size of drive.

*Other holes in drive also named as floor holes, wall holes, roof holes, stoping hole – up ward & horizontal, stoping hole – downwards hole,

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Fixation Factor I

*Different fixation factors are applied for different holes.

*Drilling bench blasting, when holes are vertical, fixation factor is 1.

*For inclined holes, it is easear to loosen toe, fixation factor is less than 1.

*In drive blasting, number of holes are blasted at one time.

*Some holes have to loose burden up ward & some down wards.

*Different fixation factors are used to include effect of multiple hole & that of gravity.

Look Out Angle I

*look out angle has to be calculated for peripheral holes.

* Look out angle depends on drill hole length.

*For 3 m hole 3 degree (5cm/m) look out angle is sufficient.

*Look out angle is very important for lifter holes (floor holes).

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Angled Cut I

*When cut holes (breaking holes) are put at an angle to be axis of working (drive) pattern is known as angle hole.

*Drive face is utilized as free face to wards which initial blasting power is directed.

*This results a cavity, towards which subsequent blasting is directed.

*Angle cut has some limitation provided below.

*Width of drive & size of drill machines with mounting limits angle related to axis of drive at which holes can be drilled.

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Angled Cut II

*Accuracy of drilling – in some of angled pattern, pair of holes drilled should meet as close as possible to promote a flash over.

*In practice, it is difficult to achieve, then by less pull is resulted.

*Dependence of blast hole depth on width of drive, hole requires slopping (inclination).

*Difficult to caller & drill hole with accuracy in desired direction.

*In angle cut, use of orientation of rock bed, available joint plane & cracks, joint pattern makes limitation.

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Angled Cut III

*This pattern is also drilled in hard & tough rock.

*This pattern results in to fly rock & high consumption of explosive.

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Wedge Cut I

*Wedge cut is also called “V” cut.

*In ‘V” cut, two holes are drilled at a horizon (1 to 1.5 m above floor) almost at center of face (width wise).

*Ideally both holes should meet at their apex, but in practice it is difficult to achieve.

*Instead of “V”, a wedge is resulted.

*Angle of subsequent holes are drilled at same horizon are increased in such a way that round holes (at sides) are at 90-95 degree to face.

*In harder strata (formation) a double or triple V or wedge cab be drilled.

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Wedge Cut II

*While blasting V holes are given initial delay& to subsequent holes are given an increasing order.

*A wedge of rock that is pulled out first is ultimately converted in to a slot / kef / slit, that has been generated by blasting.

*Rest of holes of pattern are drilled parallel to axis of drive / tunnel & blasted in a sequential order using delay detonator & taki8ng advantage of initial free face created.

*This pattern is suitable for medium hard to hard strata.

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Wedge Cut III

*An average of 45 to 50% of drive width is archived.

*Angle of cut should not be too acute & should not be less than 60 degree.

*More acute angle requires higher charge concentration in holes.

*Cut usually consists of two Vs but for deeper round Vs could be three or four.

*Holes with in each V should be given same delay number & there should be delay interval of 50ms between consecutive V to allow time for broken rock displacement & swelling..

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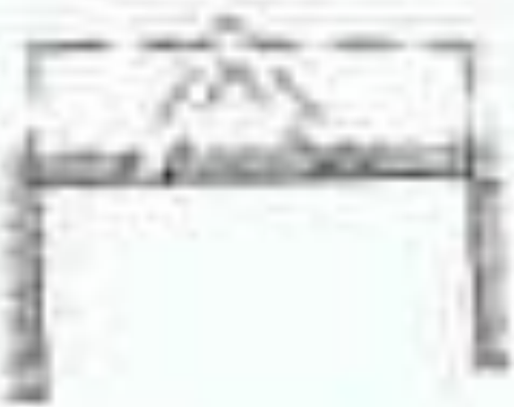
Wedge Hole I *

Elevation



Plan

(b) Wedge cut

(Isometric views)

Fan Cut I

*In this pattern holes are drilled in a fan like fashion.

*Holes are drilled at a horizon 1-1.5m above floor.

*Holes are set at different angle in an increasing order, starting from one side of face, so that large hole (at other side of face) is at 90-95 degree.

*Some time second fan is drilled at another horizon with in 0.3m above or below thos horizon.

*Holes are fired in a sequential order starting from first hole (one having least angle at one side of face).

*This results an initial ditch / kerf to wards which blasting of rest of holes can be direcred by use of delay detonators.

Fan Cut II

*This pattern is suitable for soft to medium hard strata.

*Hole director, some shot of template is used during manual drilling to achieve better accuracy of drilling.

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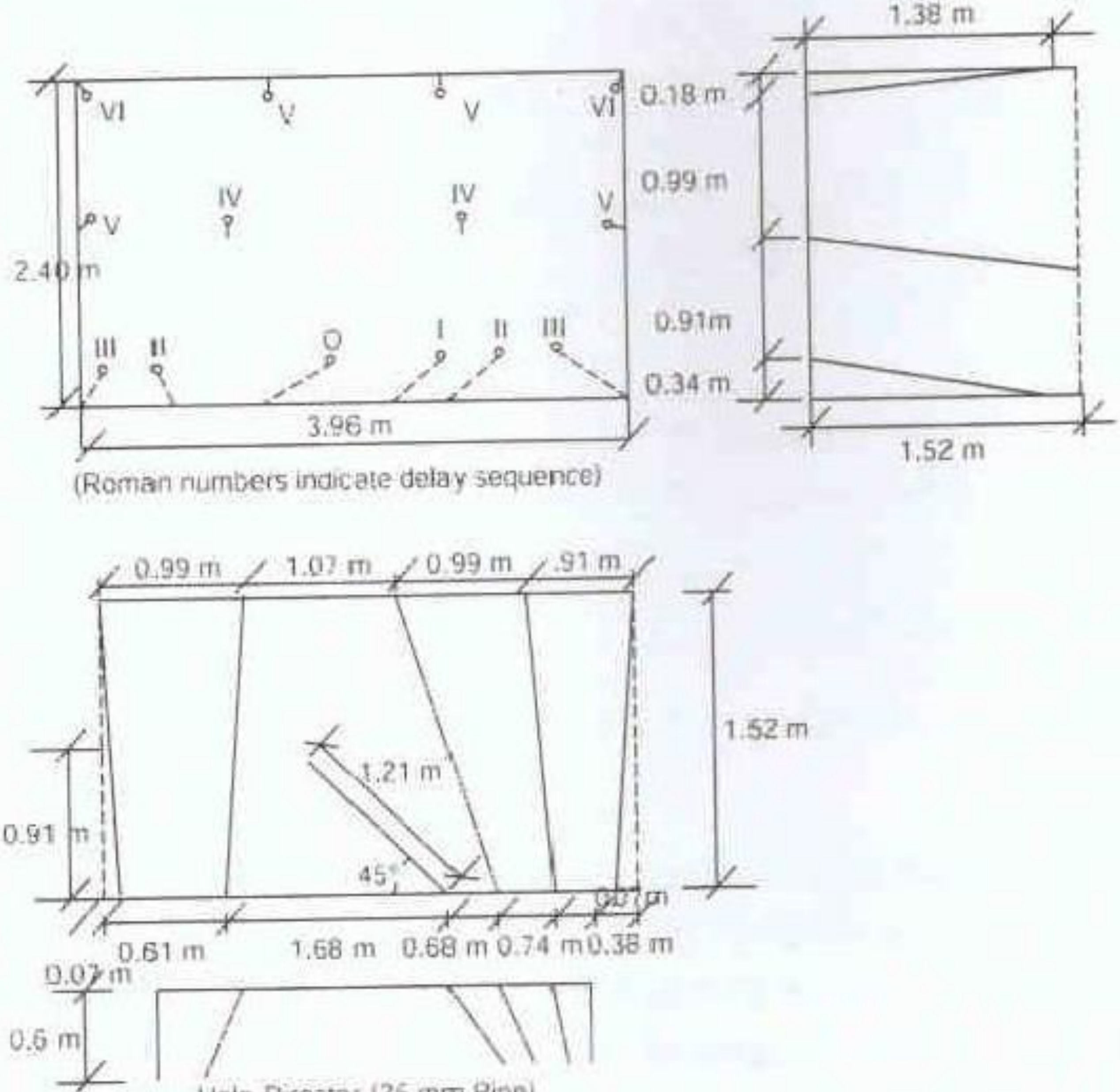
Fan Cut



(d) Blasting off the solid – fan cut (different views)

Drag Cut I

*In this pattern holes are drulled like a fan cut pattern but in a vertical plane.

*When these breaking holes (cut holes) are blasted, an under cut (slot / ditch) at face is created.

*drilling & blasting of rest of holes are directed toward these cavity.

*This pattern is suitable for soft to medium hard starta.

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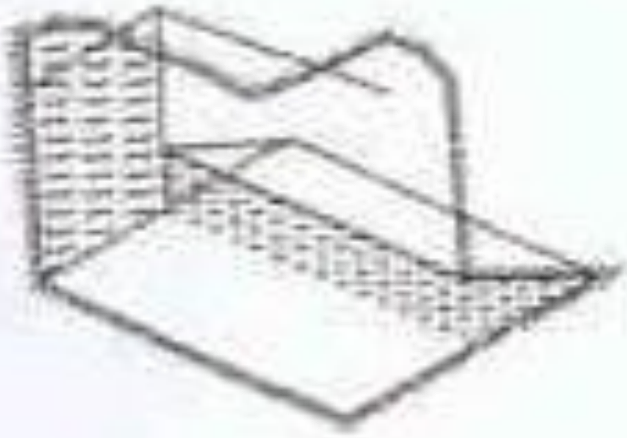
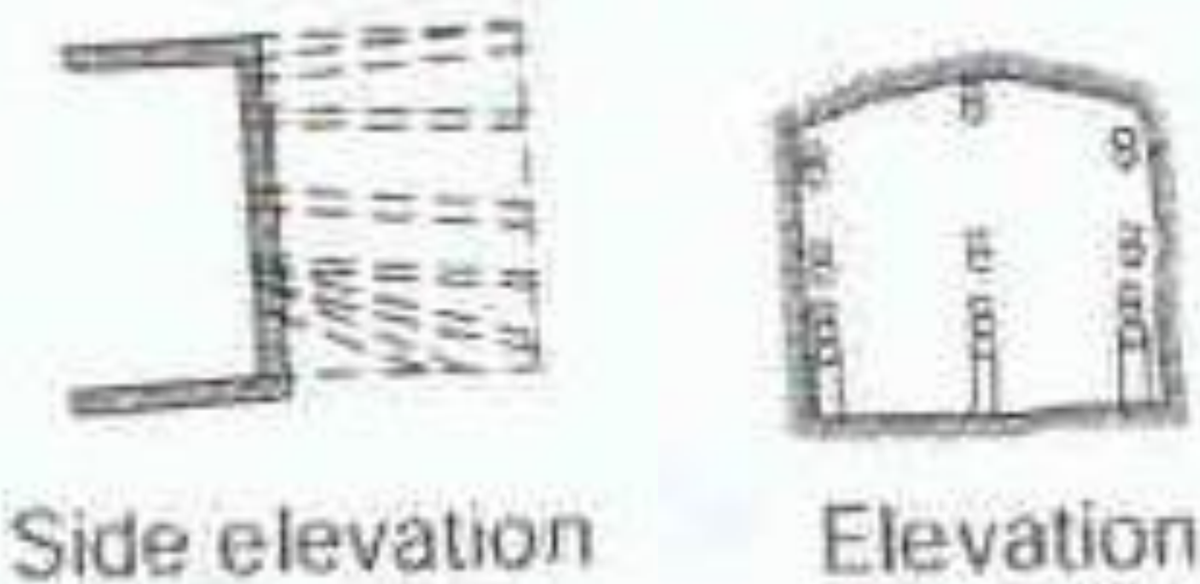
Drag Cut *



(c) Drag cut



Pyramid Cut I

*This pattern can be drilled for any drivage work – horizontal or upward.

*A cluster of holes (having 4-6 holes) is drilled in center of face directing towards a common apex so that a pyramid is formed after blasting.

*Angle of subsequent holes drilled around cut holes are increased in such a way that round holes (at sides) are 90- 95 degree to face.

*This pattern is popular during shaft sinking & raising operation & suitable for all type of strata.

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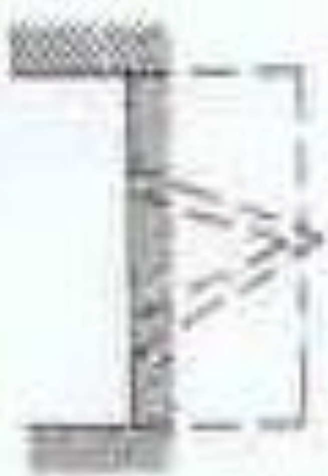
Pyramid Cut *

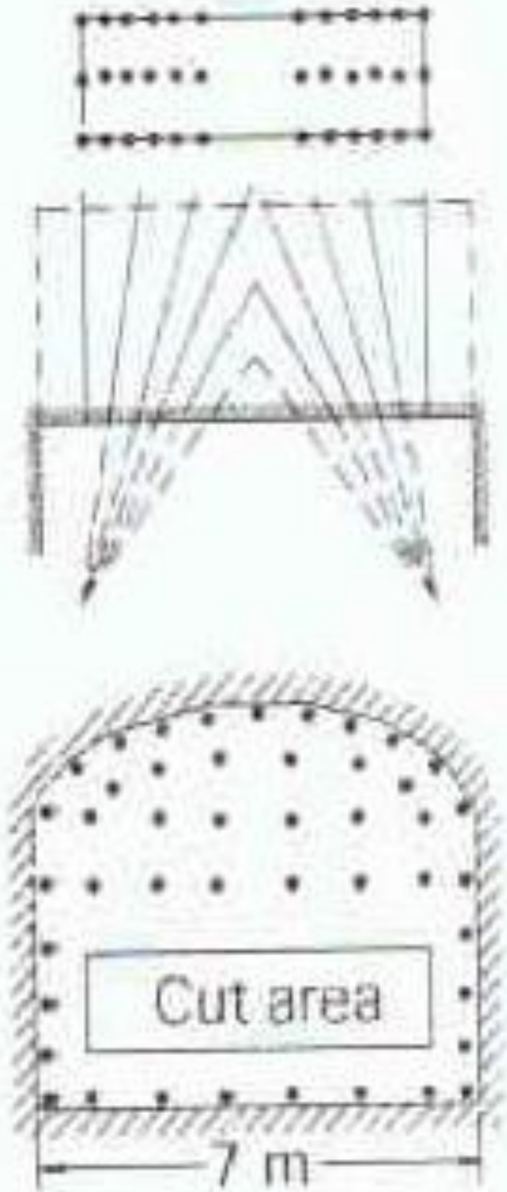
(a) Pyramid cut

Angled



Side elevation

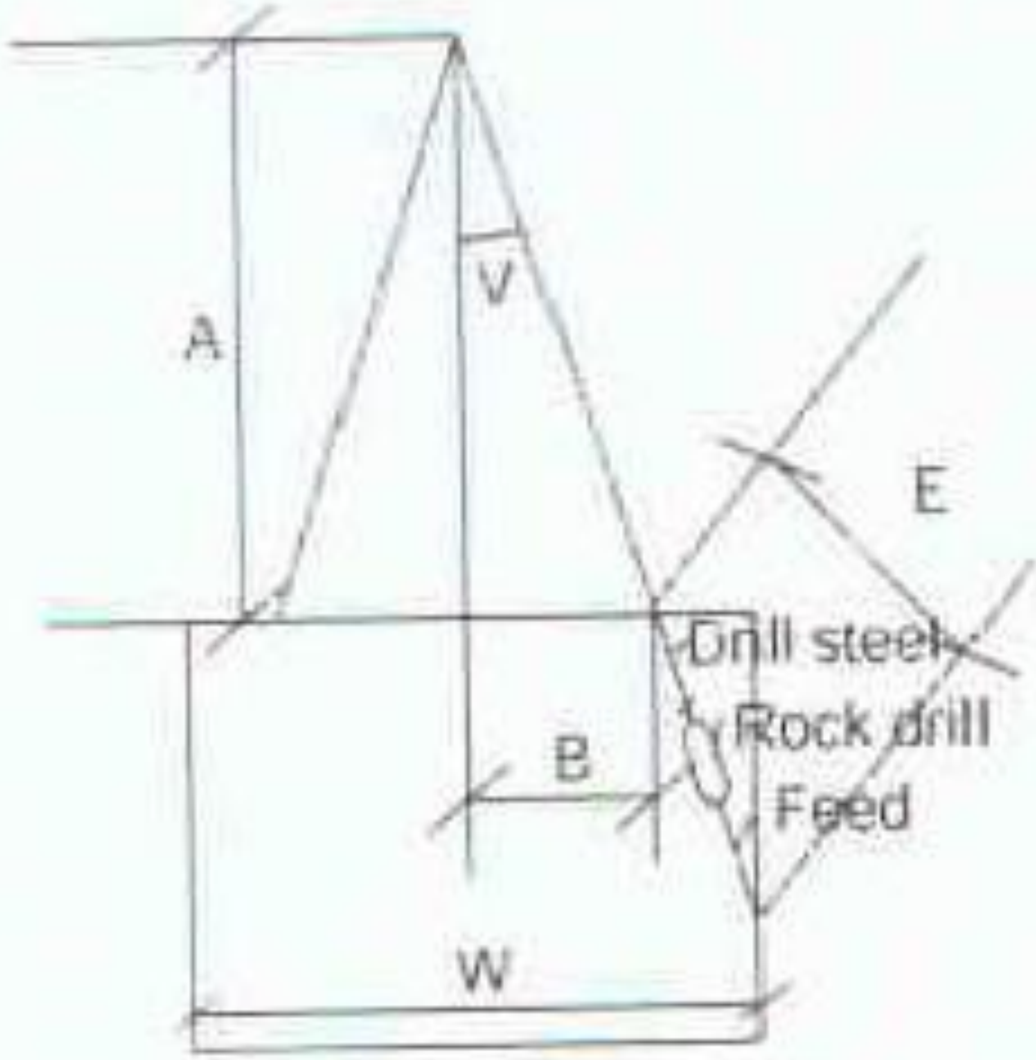
Elevation



(e) V cut

V
Cut

Angled Cut Holes



(f) Influence of tunnel's width in angled cut patterns on the advance/blast

Drilling I

*Blast holes of smaller diameter are easy to drill by hand held & pusher leg mounted drill.

*Larger diameter holes are drilled using rig or jumbo mounted with drifter.

*In small diameter holes cartridge of 25mm is used.

*In India explosive cartridge are manufactured at 25mm, 32mm, 40mm & others diameter.

*Slurry & ANFO is also used for blasting drill holes.

*In Indian condition Emulsan Explosive is most used explosive for underground mines.

*ANFO is not allowed for underground mines.

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Drilling II

*Large diameter hole drilled with help of Drill Jumbo has reduced number of hole in drilling face.

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Reference I

*Surface and Underground Excavation- Methods, Techniques and Equipment by Ratan Raj Tatiya. Page 201 to 214